POUYA AZARANDAZ

Mechanical Engineer | FEA & Structural Simulation | CAD, GD&T & DfAM

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PROFESSIONAL SUMMARY

Mechanical engineer with an M.Sc. from Politecnico di Torino combining FEA-driven structural simulation, additive manufacturing, and hands-on experimental validation, plus 2 years of industrial CAD and GD&T design. M.Sc. thesis closed the loop between ABAQUS finite element modelling, quasi-static compression testing, and CT-based defect characterisation of L-PBF lattice structures, with FE-to-experiment force correlation of $R^2 = 0.853$ and peak-force error within 5.6%. Strengths in structural analysis, lightweight design for additive manufacturing, simulation-to-test correlation, and review-ready engineering documentation. Eager to apply this combination to flight-hardware development for space.

CORE TECHNICAL SKILLS

FEA & Structural Analysis: ABAQUS (static, explicit, contact, compression, nonlinear), simulation-to-experiment correlation, mesh strategy and convergence checks, quasi-static energy-balance verification (kinetic/internal < 10%), Altair Inspire, ANSYS Mechanical (introductory).

CAD, GD&T & Mechanical Design: SolidWorks 3D part and assembly modelling, GD&T-annotated 2D manufacturing drawings, drawing interpretation, tolerance application, design reviews and revision control across 2 years of industrial practice.

Additive Manufacturing & DfAM/DfMA: Laser powder bed fusion (L-PBF), TPMS lattice and graded-density design (nTopology), overhang and support strategy, build orientation, process-parameter mapping (VED), manufacturability and post-processing planning.

Testing, Characterisation & Validation: Quasi-static compression (ISO 13314), 3-point bending and Vickers indentation per ASTM-referenced protocols, X-ray CT defect quantification (VGStudio MAX, VGDefX), SEM fractography, test-plan writing, fixture and specimen preparation, data traceability.

Materials & Lightweight Structures: AlSi10Mg, Ti-6Al-4V, NiTi, advanced ceramics, steel; lattice and thin-walled architectures for impact and energy absorption; defect-mechanics interpretation linking porosity, lack-of-fusion, and collapse behaviour.

Programming, Data & Documentation: Python, MATLAB, KNIME for engineering data analysis and supervised ML; technical reports, validation documentation, CAD drawing packages, academic manuscripts, peer review for Thin-Walled Structures (Elsevier).

PROFESSIONAL EXPERIENCE

M.Sc. Researcher — Lattice Structures, FEA & Test Correlation *Oct 2023 – Jul 2025*

Politecnico di Torino, Italy *Torino, IT*

- Designed uniform and functionally graded TPMS lattice architectures (Schwartz-P, Gyroid, FRD) in nTopology for lightweight load-bearing applications; applied DfAM constraints covering overhang angle, build orientation, support strategy, and minimum wall thickness for L-PBF AlSi10Mg.
- Built nonlinear ABAQUS/Explicit FEA models with rigid-plate contact, friction, and elastic-plastic material definition to predict stiffness, peak load, plateau response, and energy absorption; achieved force-displacement correlation of $R^2 = 0.853$ against experimental data, with peak-force error -5.55% and absorbed-energy error -3.93%.
- Ran quasi-static compression campaigns to ISO 13314 (5 specimens per configuration), wrote test procedures, prepared specimens, and used the data to validate FEA and refine design assumptions — the same simulation-to-test workflow used to qualify hardware against environmental loads.
- Quantified manufacturing-induced defects (porosity, lack-of-fusion, unfused powder) by X-ray CT in VGStudio MAX/VGDefX and linked defect content to mechanical performance via SEM fractography, supporting design margin and inspection decisions.
- Published findings at the 4th International Online Conference on Materials (MDPI); two journal manuscripts in preparation; serve as peer reviewer for Thin-Walled Structures (Elsevier).

Mechanical Design Engineer *Sep 2021 – Jul 2023*

Peyman Modiriat Saze Co., Tehran, Iran *2 years*

- Produced SolidWorks 3D parts and assemblies for industrial mechanical equipment and released GD&T-annotated 2D manufacturing drawings covering datums, tolerances, surface finishes, and assembly interfaces.
- Owned drawing packages through iterative design reviews and revision control, reducing interpretation queries between design, manufacturing, and procurement and improving release readiness.

- Coordinated cross-revision technical documentation and configuration records, applying the same traceability discipline expected for mechanical ICDs and review-ready hardware documentation.

Research Assistant — Mechanical Characterisation of Ceramics *Jul 2024 – Oct 2024*
Politecnico di Milano, Italy *Milano, IT*

- Ran 3-point bending and Vickers indentation campaigns on advanced ceramic specimens following ASTM-referenced test protocols; wrote test procedures, designed specimen preparation steps, and produced traceable test reports.
- Operated benchtop test equipment hands-on, defined fixturing for small brittle specimens, and reduced raw load-displacement data into engineering metrics (flexural strength, fracture toughness).

SELECTED PROJECTS & RESEARCH

M.Sc. Thesis — FEA & Experimental Study of L-PBF TPMS Lattices *2023 – 2025*

- Closed-loop workflow: CAD lattice design → L-PBF fabrication → X-ray CT inspection → quasi-static compression test (ISO 13314) → ABAQUS/Explicit FEA → SEM fractography. FE simulation reproduced the FRD-30 global response with R^2 = 0.853 and peak-force error within 5.6%.
- Separated designed void fraction from manufacturing-induced defect ratio (range 0.61–1.49%), showing that relative density alone is insufficient for ranking lightweight structures — topology, defects, and collapse stability must be considered together.

Crushing Analysis of Thin-Walled Tubes under Axial Impact — Published *2022*

- Co-authored a peer-reviewed study combining axial impact testing and ABAQUS FEA of a novel multibody thin-walled tube; published in the International Journal of Applied Mechanics, 14(09), 2250042.

Supervised Machine Learning for L-PBF Parameter Optimisation *2024 – 2025*

- Applied supervised ML (Python, MATLAB, KNIME) to powder-bed-fusion process datasets; manuscript on ML-based parameter optimisation in preparation.

EDUCATION

M.Sc. Mechanical Engineering *2022 – 2025*
Politecnico di Torino, Italy *Final grade 95/110*

- Thesis: A Comparative Multi-Method Study of Uniform and Graded TPMS Lattices Fabricated via Additive Manufacturing (FEA, compression testing, CT and SEM characterisation). Relevant coursework: Machine Design, Additive Manufacturing Systems & Materials, Numerical Modelling, Materials Integration & Joining Technologies, Python.

B.Sc. Mechanical Engineering *2017 – 2021*
Shahid Bahonar University of Kerman, Iran *Top 5% of department*

- Thesis: Experimental and Numerical Crushing Analysis of a Novel Multibody Thin-Walled Circular Tube under Axial Impact Loading. Merit scholarship for direct M.Sc. admission.

PUBLICATIONS & PROFESSIONAL SERVICE

- **Journal:** Mozafari, A., Azarandaz, P., Darijani, H., Shahsavari, H. (2022). Experimental and numerical crushing analysis of a novel multibody thin-walled circular tube under axial impact loading. *International Journal of Applied Mechanics, 14(09), 2250042.*
- **Conference & in preparation:** A Comparative Multi-Method Study of Uniform and Graded TPMS Lattices — 4th International Online Conference on Materials, MDPI. Two journal manuscripts in preparation (TPMS FRD lattice study; ML for L-PBF parameter optimisation).
- **Peer reviewer:** Thin-Walled Structures (Elsevier) — structural mechanics and engineering materials.

CERTIFICATIONS

- **AM & Design:** Design for Additive Manufacturing (Arizona State University, Coursera); Introduction to Crystal Plasticity Modeling in Ansys Mechanical (Ansys).
- **CAD:** SOLIDWORKS Assemblies and Exam-Level Part Modelling (Dassault Systèmes, Coursera); SOLIDWORKS: Advanced Engineering Drawings (LinkedIn Learning).
- **Programming & ML:** MATLAB Onramp, Deep Learning Onramp, Regression with Deep Learning (MathWorks); Supervised Machine Learning: Regression and Classification, Neural Networks and Deep Learning (DeepLearning.AI, Coursera).

AWARDS & LANGUAGES

- **Awards:** Top 5% of graduates, Mechanical Engineering, Shahid Bahonar University of Kerman; merit-based M.Sc. scholarship (direct admission without national entrance examination). **Languages:** Persian (native), English C1 (professional), Italian A2 (developing).